

GUJARAT COUNCIL ON SCIENCE AND TECHNOLOGY
Dept. of Science and Technology, Govt. of Gujarat

ROBOFEST- GUJARAT 6.0

Guidelines for submission of Ideation Proposal
Robofest Hackathon 2026: AI-Powered Robotics Innovation Challenge

Overview:

Hackathon 2026 is an innovation-driven robotics hackathon aimed at bringing together young innovators and students to develop next-generation intelligent robotic solutions for addressing real-world societal, industrial, environmental, and scientific challenges, especially for India. The challenge encourages participants to integrate Artificial Intelligence (AI), robotics, autonomous systems, digital twin, and simulation technologies to design smart, efficient, and impactful solutions alongwith working-level Proof of Concept (PoC) model.

Participants may demonstrate their innovations through any one (or more) metric(s) or all the metrics mentioned below:

[i] functional robotic prototype; [ii] digital twin model; [iii] simulation-based system; [iv] real-time animation; [v] a hybrid physical-virtual implementation.

The open-ended nature of the challenge promotes creativity, interdisciplinary collaboration, hands-on learning, and technological innovation, fostering a future-ready workforce capable of developing affordable, scalable, and sustainable robotic solutions for the society.

Key Features:

- Open-ended real-world problem identification and development of realizable solution
- Flexibility to demonstrate solutions through working prototypes, digital twins, simulations, real-time animation or hybrid systems
- Integration of emerging technologies such as AI/ML, Reinforcement Learning, Artificial Neural Network, Computer Vision, Autonomous Systems, IoT, Embedded Systems, and Digital Twin
- Innovation-driven approach focusing on creativity and practical implementation
- Emphasis on affordable, scalable, sustainable, and socially impactful solutions

Structure:

Phase 1: Pre-Hackathon (Online Screening & Team Selection)

Teams shall submit their innovative ideas through the online application portal including:

- **Problem Identification:** Clearly define the real-world societal, industrial, environmental, or scientific challenge to be addressed.
- **Concept Proposal:** Describe the proposed AI-powered robotic solution, including the technical approach, innovation, expected functionality, and anticipated impact. The proposed solution may be demonstrated through a working prototype, digital twin, simulation-based model, real-time animation or a hybrid approach. However, the Integration of Hardware with Digital Twin/ Simulation (Hybrid Approach) shall carry maximum marks.
- **Team Profile:** Details of team members, academic background, skills, and relevant experience.

Evaluation and Shortlisting Criteria

- Innovation and originality of the proposed solution.
- Relevance of the identified problem and its potential societal/industrial impact.

- Technical feasibility, design approach, scalability, and implementation potential.

Selected teams will be shortlisted and invited to participate in the Grand On-site Robothon.

Phase 2: Grand Hackathon (36 Hours On-site Development & Demonstration)

Kick-off & Orientation

- Introduction to the Hackathon theme, challenge areas, rules, evaluation criteria, and available facilities/resources.
- Interaction with mentors and domain experts for idea refinement, technical guidance, and implementation strategy.

Development Sprint

- Teams shall undertake system design, programming, AI/ML model development, hardware integration, simulation modelling, animation, digital twin creation, and solution implementation as per their proposed concept.
- Continuous mentoring support will be provided by experts from academia, research organizations, and industry.
- On request, the teams may be allowed to continue working overnight. However, GUJCOST will reserve the right for approving such plea by the team(s) depending upon the logistics available at the venue. GUJCOST will not take any responsibility for any eventual loss of item / hardware / software under the possession of the team(s) during overnight work and/or personal injury or unethical incidence. Team-members, alongwith their mentor, have to sign a declaration form a-priori for availing night-time working schedule.
- Teams should note that CAD model, trainer model (if any), hardware component(s), miscellaneous electrical hardware including extension board / spike guard / additional power cord/ cable/ socket/ connectors, manufactured component(s), hardware sub-assembly or any program-code, may be brought in pre-built form and must be declared before the start of the event.
- Teams must possess all usual hardware accessories, mechanical tool-set, jigs & fixtures (if any), screws, bolts, nuts, threads etc.
- Teams must do the following work-modules on-site: a] all instrumentation; b] integration of motor / driver / actuator & sensor(s); c] all sub-assemblies / full assembly; d] compilation / activation of the final-level program-code, Graphical User Interface (GUI), Human-Machine Interface (HMI) and/or software –as applicable; e] completion of the full prototype; f] Working / operation of the full & final prototype.
- remote team members or off-site help are not permitted,
- internet access will be provided
- pre-trained models are allowed

Progress Review & Optimization

- Periodic review checkpoints with mentors and jury members to evaluate progress and provide technical feedback. Additional on-site mentoring support to the teams will be allowed.
- Teams will refine their solutions through testing, debugging, optimization, and performance improvement.

Final Demonstration & Submission

Teams shall submit and present:

- A functional robotic prototype, digital twin, simulation-based solution, real-time animation or hybrid physical-virtual system.

- Technical presentation covering the problem statement, methodology, system architecture, innovation, and expected impact.
- Demonstration video and technical documentation (if applicable).

Participating Eligibility

The Hackathon is open for students of Class 9 to 12 as well as students pursuing ITI, Diploma, Under-Graduate and Post-Graduate courses (Full-time) in STEM field. Students from schools, diploma institutions, colleges, universities, and other recognized educational institutes are eligible to participate.

Team Composition

- **Team Size:** A total of 5 members comprising of 4 student members with **one faculty mentor** from the same academic institution.
- Multidisciplinary teams comprising students from mechanical, electronics, electrical, computer science, AI, data science, and other aligned disciplines are encouraged.

Problem Statement

“Develop an innovative AI-powered robotic solution that addresses a real-world societal, industrial, environmental, or scientific challenge and contributes towards improving quality of life, sustainability, safety, productivity, accessibility, or efficiency in society.”

Participants are encouraged to identify challenges from any domain and develop intelligent robotic solutions capable of perception, learning, decision-making, autonomous operation, or human–robot collaboration through the integration of Artificial Intelligence and advanced robotics technologies.

The proposed solutions may be developed and demonstrated through:

- A functional hardware-based robotic prototype (rigid / compliant / flexible or soft structure)
- A digital twin or simulation-based robotic model
- A hybrid system integrating both physical and virtual environments

The solutions may focus on, but are not limited to, the following areas:

- Healthcare and Assistive Robotics
- Agriculture and precision farming
- Rescue, Disaster management and public safety
- Environmental conservation and sustainability
- Inspection, Surveillance and Predictive Maintenance
- Smart cities and infrastructure monitoring through Web-mediated Robotics
- Industrial automation and human–robot collaboration
- Education and inclusive technologies
- Autonomous mobility including ground, aerial, and underwater robotics
- Space, ocean, and extreme environment exploration
- Any other innovative societal, industrial, or scientific challenge

The proposed solutions may incorporate, but are not limited to, technologies such as Artificial Intelligence and Machine Learning (AI/ML), Deep Reinforcement Learning(DRL), Computer Vision, Generative AI, Intelligent Agents, Artificial Neural Network & Convolutional Neural Network (ANN & CNN), Natural Language Processing (NLP), autonomous navigation, sensor fusion, embedded systems, IoT, edge/cloud computing, digital twin technologies, advanced simulation environments, swarm robotics, and intelligent control systems.

The challenge is intentionally kept open-ended to encourage creativity, interdisciplinary collaboration, and technological innovation. Participants are encouraged to develop impactful, scalable, and sustainable AI-powered robotic solutions that bridge the gap between virtual design and real-world deployment, contributing towards a smarter, safer, and more sustainable future for India.

Tools & Platforms (Recommended):

Participants may use suitable tools, software platforms, open-source resources, programming languages, frameworks, and development environments for designing, developing, simulating, animating and demonstrating their AI-powered robotic solutions.

The suggested categories include, but are not limited to:

- Artificial Intelligence, Machine Learning, and Data Analytics frameworks
- Computer Vision and Intelligent Perception tools
- Robot Operating Systems (ROS) and robotic middleware platforms
- Robotics simulation, digital twin, and virtual testing environments
- 3D modelling, Computer-Aided Design (CAD), and engineering design tools
- Programming languages, software development environments, and algorithm development tools
- Embedded systems, microcontroller, and hardware development platforms
- Sensor interfacing, control systems, and autonomous navigation tools
- Internet of Things (IoT), edge computing, cloud computing, and communication platforms
- Data visualization, monitoring, and human–robot interaction development tools

Note: The above list is indicative and not exhaustive. Participants are free to use any appropriate open-source or commercial software, platforms, libraries, hardware, and tools as per their project requirements. All software, datasets, AI models, and third-party resources used must comply with applicable licensing agreements, copyright regulations, and intellectual property rights. Participants shall be solely responsible for ensuring legal and authorized use of such resources.

Deliverables:

Each participating team shall develop and submit an innovative AI-powered robotic solution addressing a real-world societal, industrial, environmental, or scientific challenge at the designated date and location. The proposed solution may be demonstrated through a functional robotic prototype, digital twin model, simulation-based system, or a hybrid physical-virtual implementation.

The final submission shall include:

- **Working Solution:** A functional robotic prototype, digital twin, simulation and/or animation model, or hybrid system demonstrating the proposed concept and its operational capabilities.
- **AI & Intelligent System Component:** Integration of AI models, algorithms, autonomous decision-making, computer vision, perception, control logic, or other intelligent functionalities as applicable to the proposed solution.
- **Technical Documentation:** A concise report covering the problem statement, system architecture, design methodology, technologies used, implementation approach, achieved outcomes, and potential societal/industrial impact.
- **Final Pitch & Demonstration:** A 5-minute presentation and live demonstration showcasing the identified problem, innovation, technical approach, developed solution, results, and future scalability.

Infrastructure Support

The following facilities shall be provided by the organizers throughout the hackathon:

- Dedicated workstation with table and chairs for each team.
- Continuous power supply with 230 V AC power sockets (minimum two sockets per team).
- High-speed Wi-Fi internet connectivity for software development and documentation.
- Basic hand tools and shared workshop facilities, including screwdrivers, pliers, wire strippers, drills, and soldering stations.
- Limited access to fabrication facilities (subject to the event schedule), including 3D printers where available.
- First-aid and emergency support.

The detailed list of facilities and equipment available at the venue shall be published on the official Robofest Gujarat website at least **15 days before the commencement of the hackathon**.

Equipment to be Brought by Teams

Each participating team shall bring its own:

- Laptop(s) and chargers
- Robot, drone, or prototype hardware
- Electronic components and sensors
- Batteries and approved chargers
- Development boards and programming cables
- Consumables required for prototype development
- Personal protective equipment (PPE), prescribed medicines wherever applicable

The organizers shall not be responsible for the availability of specialized components or tools.

Safety Guidelines

1. Only commercially manufactured Li-ion or LiPo batteries in good condition shall be permitted. Damaged, swollen, or leaking batteries shall not be allowed.
2. Battery charging shall be carried out only in the designated charging area using manufacturer-approved chargers. Batteries shall not be left unattended while charging.
3. Soldering, drilling, cutting, or use of power tools shall be permitted only in designated work areas under the supervision of the organizers.
4. The use of open flames, combustible fuels, compressed gases, pyrotechnics, hazardous chemicals, or explosive materials is strictly prohibited.
5. Indoor flight of drones shall **not** be permitted except during designated testing sessions in the approved flight arena under the supervision of the organizing committee.
6. All participants shall comply with electrical safety practices and immediately report any unsafe condition, equipment malfunction, or accident to the organizers.
7. Participants below 18 years of age shall remain under the supervision of their designated mentor or accompanying faculty member throughout the event.
8. Basic first-aid facilities shall be available at the venue. In case of an emergency, the decision of the organizers regarding continuation or suspension of activities shall be final.
9. Each team shall be solely responsible for the safety of its members, equipment, and prototype. GUJCOST and the organizers shall not be liable for any loss, damage, injury, or accident arising from the use of participant-owned equipment.

10. All participants shall comply with the instructions issued by the organizers and the Technical Advisory Committee (TAC). Failure to adhere to the safety guidelines may result in disqualification from the competition.

Evaluation Criteria:

Evaluation Parameter	Weightage (%)
Problem Relevance, Innovation & Societal/Industrial Impact	20%
AI Integration & Intelligent Functionality	15%
System Design, Technical Approach & Engineering Excellence	15%
Hardware Prototype Development & Performance	25%
Digital Twin/Simulation / Animation Model Accuracy & Performance	15%
Integration of Hardware with Digital Twin/Simulation (Hybrid Approach)	5%
Technical Presentation, Demonstration & Documentation	5%

This Hackathon will foster innovation and interdisciplinary collaboration by encouraging participants to develop impactful AI and robotics-based solutions for real-world challenges. It will enhance practical skills in emerging technologies such as artificial intelligence, autonomous systems, and digital twin technologies. The event will promote a culture of research, innovation, and entrepreneurship, leading to scalable, sustainable, and socially beneficial technological solutions.

Note: The guidelines related to the eligibility, timelines, formats of submissions, prizes and conduct are published on the Robofest Gujarat Webportal and those shall be binding to all participating teams.